

Recovery of bandlimited graph signals based on the reproducing kernel Hilbert space

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ARTICLE INFO

Keywords:

Signal recovery on graphs
Unconstrained optimization model
Reproducing kernel Hilbert space
Bandlimited graph signal
Representer theorem

ABSTRACT

Signal recovery on graphs is attracting more and more attentions. Based on the smoothness assumption, the signal recovery problem can be formulated as an unconstrained optimization model. Although the model can be solved analytically, the computational cost is too expensive. In this paper, to reduce the computational complexity, we skillfully construct a kernel function, such that the smoothness item is equivalent to the norm of the signal modulo a constant in the reproducing kernel Hilbert space. Based on this, by using the Representer Theorem of reproducing kernel Hilbert space, the optimization problem for recovering the signal from a given bandlimited space, can be solved in a lower-dimensional subspace corresponding to the known labeled set. Experiments for the recovery of artificial graph signals in GSPBOX and real data in the recommendation system show very encouraging results.

1. Introduction

With the explosive growth of information and communication, datasets are being generated at an unprecedented rate. At the same time, datasets often present characteristic structures or are defined on irregular domains. In recent years, the increasing interesting for signal processing for big data or irregular data has resulted an emerging field of signal processing on graphs. Many problems can be naturally converted to signal processing models in terms of graphs, such as social and economic networks [1–3], customer preferences for service or goods [4], brain studies [5,6], learning algorithms [7,8] and many others.

Unlike time-series and images, a graph signal is modeled as a scalar function defined on the vertices of a graph. The data is expected to be influenced by similarities between vertices. Exploiting the underlying graph topology, many researchers extend graph signal processing on various applications, including graph signal filtering [9–14], graph wavelets [15–17], graph topology learning [18–20], graph vertex-frequency analysis [21–24], graph Fourier transform [25–27], uncer-

tainty principle [28], and graph signal sampling [29–41]. For a recent review, the readers are referred to [42].

This paper concerns on the problem of signal recovery on graphs. Let y be a signal defined on the vertex set $\mathcal{V} := \{v_1, \dots, v_N\}$ of an undirected, connected, weighted graph. Assume only a part of values of y , $\{y(v) | v \in \mathcal{V}_0\}$, are known, where $\mathcal{V}_0$ is a proper subset of $\mathcal{V}$. Our goal is to recover those unknown values of y on $\mathcal{V} \setminus \mathcal{V}_0$. This problem can be formulated mathematically as follow: Given a set $\mathcal{V}_0 \subset \mathcal{V}$ and a function $y : \mathcal{V}_0 \rightarrow \mathbb{R}$, finding a function

$$f : \mathcal{V} \rightarrow \mathbb{R} \text{ such that } f(v) = y(v) \text{ for all } v \in \mathcal{V}_0. \quad (1.1)$$

Since $\mathcal{V}_0$ is a proper subset of $\mathcal{V}$, there are infinitely many functions f that satisfy the above conditions. It is a very interesting question to choose the ‘best’ one according to practical applications. To do this, a non-negative index $\sigma(f)$ is introduced to measure the ‘goodness’ or ‘badness’ of f . Usually, smaller $\sigma(f)$ means better prediction for f . By means of $\sigma(f)$, the problem (1.1) can be modified as the following optimization problem:

$$\min_{f \in \mathbb{R}^{\mathcal{V}}} \sigma(f) \quad \text{subject to} \quad f(v) = y(v), \quad \forall v \in \mathcal{V}_0. \quad (1.2)$$

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where $\mathbb{R}^{\mathcal{V}}$ denotes the vector space consisting of all the functions $f : \mathcal{V} \rightarrow \mathbb{R}$.

Problem (1.2) is a constrained optimization problem, which is usually not easy to solve. Meanwhile actual observations are often corrupted by noises. Therefore, we consider the following unconstrained optimization problem:

$$\min_{f \in \mathbb{R}^{\mathcal{V}}} \sum_{v \in \mathcal{V}_0} |f(v) - y(v)|^2 + \gamma \sigma(f), \quad (1.3)$$

where $\gamma > 0$ is a regularization parameter used to balance the weight between the error $\sum_{v \in \mathcal{V}_0} |f(v) - y(v)|^2$ and the quality index $\sigma(f)$.

Although the solution of problem (1.3) may not exactly satisfy the constraint conditions $f(v) = y(v)$, $\forall v \in \mathcal{V}_0$, it has advantages over the problem (1.2) in at least two aspects: (a) It is an unconstrained problem which can be solved more easily; (b) When $\gamma > 0$, such a compromise on the regularization term allows certain degree of approximation error, but brings the benefit of an increase of signal quality, which often makes the solution have better physics in applications.

This problem has been studied extensively. Some problems from different disciplines are ultimately attributed to this problem. For example, (a) when $\mathcal{V}$ consists of N points equidistant on the temporal axis, the function defined on $\mathcal{V}$ is the classical discrete time signal. In this case, the problem is the classical sampling recovery problem [30,31,33,35]. Similarly, if $\mathcal{V}$ is set to be the coordinates of a digital image, the problem is the classical problem of image recovery or inpainting [43,44]; (b) if $\mathcal{V}$ is the set of entries of a matrix f and $\sigma(f)$ is set to be the rank of the matrix, then the problem refers to the problem of low-rank matrix recovery from incomplete observations [45]; (c) let $\mathcal{V} := \{(i, j) | i = 1, \dots, M; j = 1, \dots, N\}$ and denote $r(i, j)$ the rating of the i th user to the j th item, if only a few part of ratings $\{r(i, j)\}_{(i, j) \in \mathcal{V}_0}$ are known, then the problem is the prediction model in the recommendation system [31,46].

The measurement $\sigma(f)$ should be chosen according to practical applications. Several approaches have been used to measure the non-negative index $\sigma(f)$. In [44,47,48], the index $\sigma(f)$ is quantified in terms of the total variation, then the problem (1.2) is called as the problem of graph signal inpainting via total variation minimization. In Tikhonov regularization [49,50], $\sigma(f)$ is specified as the smoothness measurement by using the graph Laplacian. Generally, by properly constructing the adjacent matrix $\mathbf{W}$ of an undirected, connected, weighted graph $\mathcal{G} = (\mathcal{V}, \mathbf{W})$, one can use the smoothness functional

$$\sigma(f) = \sum_{i=1}^N \sum_{j=1}^N W(i, j) |f(v_i) - f(v_j)|^2 \quad (1.4)$$

to measure the ‘goodness’ of the function $f : \mathcal{V} \rightarrow \mathbb{R}$ quantitatively. It is known that the computational complexity of solving the problem (1.3) is $O(N^3)$, where N is the cardinality of the vertex set $\mathcal{V}$.

The computational complexity $O(N^3)$ for solving the problem (1.3) is too expensive in some applications. For example, in the recommendation system mentioned above, the numbers of users M and items N are very large, and one needs to recover the rating matrix $r(i, j)$ for $i = 1, \dots, M$ and $j = 1, \dots, N$ from a very sparse ratings $\{r(i, j)\}_{(i, j) \in \mathcal{V}_0}$. With this model it needs to solve the problem (1.3) for each $i \in \{1, \dots, M\}$ and the computational complexity $O(MN^3)$ is too large. It is a very challenging problem to find a new strategy to solve the problem (1.3) more efficiently.

In this paper, we focus on the following *bandlimited graph signal recovery problem* (BGSRP), which is more general than the problem (1.3):

$$\text{BGSRP : } \min_{f \in \mathcal{F}} \sum_{v \in \mathcal{V}_0} |f(v) - y(v)|^2 + \gamma \sigma(f), \quad (1.5)$$

where $\sigma(f)$ is defined by (1.4), and $\mathcal{F} := \text{BL}_n(\mathcal{G})$ is the n -dimensional bandlimited subspace of $\mathbb{R}^{\mathcal{V}}$ for a given $1 \leq n \leq N$. When $n = N$, we have $\mathcal{F} = \mathbb{R}^{\mathcal{V}}$ and the problem (1.5) degenerates into the problem (1.3). It is proved that, under the condition that the first n eigenvalues and

eigenvectors of the graph Laplacian are known, the analytic solution of problem (1.5) can be obtained with the computational complexity $O(Nn + (n + \ell)n^2)$, where ℓ is the cardinality of $\mathcal{V}_0$. By skillfully utilizing the theory of reproducing kernel Hilbert space, we design a new algorithm to solve the problem (1.5) with the computational complexity $O(Nn + (n + \ell)\ell^2)$, which is less than the complexity of the analytic solution provided that $\ell \leq n$. Particularly, if $N + \ell^2 = o(n^2)$, the higher order infinitesimal of n^2 , then our proposed algorithm is much faster than the computation of the analytic solution.

1.1. Contributions

The contributions of this paper are listed as follows:

- A general optimization model, bandlimited graph signal recovery problem (BGSRP), is summarized for the signal recovery on undirected, connected, weighted graphs. One prior is imposed, i.e. the graph Fourier coefficients of the signal should be equal to 0 after the bandwidth. The model can maintain both computational efficiency and certain physical properties.
- To reduce the computational cost, we skillfully construct a kernel and the corresponding reproducing kernel Hilbert space. With this kernel, the smoothness item is equivalent to the norm of the signal modulo a constant and thus the BGSRP can be solved in the reproducing kernel Hilbert space. Accordingly, by using the Representer Theorem, the BGSRP can be solved in a lower-dimensional subspace corresponding to the known labeled set. When the eigendecomposition of graph Laplacian is known a prior, our algorithm reduces the computational cost greatly.

1.2. Outline of the paper

The rest of the paper is organized as follows: In Section 2, a brief introduction to graph Fourier transforms is stated. As a core tool that will be utilized in this paper, the basic concepts of the reproducing kernel Hilbert space and the Representer Theorem are stated. In Section 3, the analytic solution of the BGSRP is deduced. By skillfully constructing a kernel function, the BGSRP is formulated as an optimization problem in a reproducing kernel Hilbert space and accordingly can be solved equivalently in a lower-dimensional subspace according to the Representer Theorem. The computational complexity is also presented. Finally, in Section 4, experiments are conducted.

1.3. Notations

Throughout this paper we use the following notations: Matrices and vectors are represented by uppercase and lowercase boldface letters, respectively. The entries of a matrix $\mathbf{A}$ are denoted as $A(i, j)$ or A_{ij} , and the entries of a vector $\mathbf{x}$ are denoted as $x(k)$ or x_k . The transpose of the matrix $\mathbf{A}$ is represented by $\mathbf{A}^T$. For a vector $\mathbf{x}$, the notation $\text{diag}(\mathbf{x})$ represents a diagonal matrix with entries of the vector $\mathbf{x}$ along the main diagonal, and $\|\mathbf{x}\|_2 := \sqrt{\mathbf{x}^T \mathbf{x}}$ is the Euclidean norm of $\mathbf{x}$. For a positive integer N , the notations $\mathbf{1}_{N \times N}$ and $\mathbf{I}_N$ denote respectively the matrix and vector of order N whose entries are all ones, and $\mathbf{I}_N$ is the identity matrix of order N . The dimension of a subspace $\mathcal{H}$ in $\mathbb{R}^N$ is denoted by $\dim(\mathcal{H})$. The direct sum of two subspaces $\mathcal{H}_1$ and $\mathcal{H}_2$ is denoted as $\mathcal{H}_1 \oplus \mathcal{H}_2$. The notation $\text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ stands for the space spanned by the vectors $\mathbf{u}_1, \dots, \mathbf{u}_k$.

2. Preliminaries

2.1. Fourier transforms for graph signals

We briefly review basic concepts of graph signal processing. More details can be found in [51].

Consider an undirected, connected, weighted graph $\mathcal{G} := (\mathcal{V}, \mathbf{W})$ with the vertex set $\mathcal{V} := \{v_1, \dots, v_N\}$ and the weighted adjacency matrix $\mathbf{W}$ whose entry $W(i, j)$ reflects the similarity between two vertices v_i and v_j . The graph Laplacian is defined as $\mathbf{L} := \mathbf{D} - \mathbf{W}$, where $\mathbf{D}$ is the diagonal degree matrix with the diagonal element $d_i := \sum_{j=1}^N W(i, j)$ for $i = 1, \dots, N$. The graph Laplacian $\mathbf{L}$ is widely used as a basic building block to develop new techniques for graph signal processing. Since $\mathbf{L}$ is a real symmetric and positive semi-definite matrix, it has the eigen-decomposition

$$\mathbf{L} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T, \quad (2.6)$$

where $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_N)$ is a diagonal eigenvalue matrix, $\mathbf{U} = (\mathbf{u}_1, \dots, \mathbf{u}_N)$ is an orthonormal matrix whose columns $\{\mathbf{u}_k\}_{k=1}^N$ are the eigenvectors of $\mathbf{L}$ and the first column $\mathbf{u}_1 = \mathbf{1}_N / \sqrt{N}$ is the unit vector with equal components. Without loss of generality, we assume that the real, non-negative eigenvalues are ordered as $0 = \lambda_1 < \lambda_2 \leq \lambda_3 \leq \dots \leq \lambda_N$. It is easy to see that $\mathbf{L}\mathbf{u}_i = \lambda_i \mathbf{u}_i$ for $i = 1, \dots, N$.

A graph signal is a function defined on the vertex set, i.e., $f : \mathcal{V} \rightarrow \mathbb{R}$. It is easy to see that f can be equivalently represented as the N -dimensional column vector $\mathbf{f} := (f(v_1), \dots, f(v_N))^T \in \mathbb{R}^N$, where $f(v_i)$ indicates the value of f on the vertex v_i . Conventionally, if the value of $f(v_i)$ is known, we call $f(v_i)$ the label of v_i and the vertex v_i is labeled.

The classical inverse Fourier transform is defined as the expansion of a function f in terms of the eigenfunctions e^{ikx} of the Laplace operator, that is, $f = \sum_k c_k(f) e^{ikx}$, where the coefficients $\{c_k(f)\}$ are called the Fourier transform of f . Analogously, the inverse Fourier transform of a graph signal $f : \mathcal{V} \rightarrow \mathbb{R}$ is defined as the expansion of the vector $\mathbf{f} \in \mathbb{R}^N$ in terms of $\{\mathbf{u}_k\}_{k=1}^N$, that is

$$\mathbf{f} = \sum_{k=1}^N \hat{f}_k \mathbf{u}_k = \mathbf{U} \hat{\mathbf{f}}, \quad (2.7)$$

where the vector of coefficients

$$\hat{\mathbf{f}} := (\hat{f}_1, \dots, \hat{f}_N)^T = \mathbf{U}^{-1} \mathbf{f} \quad (2.8)$$

is called the Fourier transform of the graph signal f .

Let us consider the index of oscillation or smoothness for a graph signal f :

$$\sigma(f) := \sum_{i=1}^N \sum_{j=1}^N W(i, j) |f(v_i) - f(v_j)|^2.$$

It is easy to understand that the larger the value $\sigma(f)$ is, the more intensively the signal oscillates and the worse its smoothness is. Since $\sigma(f) = \mathbf{f}^T \mathbf{L} \mathbf{f}$ for any function $f : \mathcal{V} \rightarrow \mathbb{R}$ and $\mathbf{L}\mathbf{u}_k = \lambda_k \mathbf{u}_k$, $k = 1, \dots, N$, we have that $\sigma(\mathbf{u}_k) = \lambda_k$, $k = 1, \dots, N$, which implies that from $\mathbf{u}_1$ to $\mathbf{u}_N$, the frequency is getting higher and higher. For $1 \leq n \leq N$, we call

$$\mathbf{BL}_n(\mathcal{G}) := \{f \mid \hat{f}_{n+1} = \dots = \hat{f}_N = 0\} \quad (2.9)$$

the bandlimited space with the highest frequency λ_n . It is easy to see that a graph signal $f : \mathcal{V} \rightarrow \mathbb{R}$ is in $\mathbf{BL}_n(\mathcal{G})$ if and only if $\mathbf{f} = \sum_{k=1}^n \hat{f}_k \mathbf{u}_k$.

2.2. The reproducing kernel Hilbert space

The reproducing kernel Hilbert space (RKHS) is an applied mathematical theory which plays an important role in machine learning. An RKHS is determined by a bivariate kernel function defined on a given set. Let $\mathcal{X}$ be a non-empty set, the function $\kappa : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is said to be a kernel function if it is symmetric, i.e. $\kappa(x, y) = \kappa(y, x)$, and satisfies the following positive definite condition,

$$\sum_{i=1}^n \sum_{j=1}^n c_i c_j \kappa(x_i, x_j) \geq 0, \quad \forall x_i \in \mathcal{X}, c_i \in \mathbb{R}, i = 1, \dots, n. \quad (2.10)$$

With a kernel function $\kappa : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$, a Hilbert space $\mathcal{H}$ consisting of some functions $f : \mathcal{X} \rightarrow \mathbb{R}$ can be defined uniquely to satisfy the following reproducing kernel property:

$$f(x) = \langle f, \kappa(\cdot, x) \rangle_{\mathcal{H}}, \quad \forall f \in \mathcal{H}, x \in \mathcal{X}, \quad (2.11)$$

where $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ is the inner product of the space $\mathcal{H}$. Owing to the property (2.11), the space $\mathcal{H}$ is called the reproducing kernel Hilbert space and κ is called the corresponding reproducing kernel.

As a special case, if $\mathcal{X} = \{v_1, \dots, v_N\}$ is a finite set, then the kernel function $\kappa : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ is determined uniquely by the following symmetric and positive semi-definite matrix of order N :

$$\mathbf{K} = \begin{bmatrix} \kappa(v_1, v_1) & \dots & \kappa(v_1, v_N) \\ \vdots & & \vdots \\ \kappa(v_N, v_1) & \dots & \kappa(v_N, v_N) \end{bmatrix}. \quad (2.12)$$

In this case, $\mathcal{H}$ consists of all the linear combination of $\kappa(\cdot, v_i)$, $i = 1, \dots, N$, that is

$$\mathcal{H} = \text{span}\{\kappa(\cdot, v_i)\}_{i=1}^N. \quad (2.13)$$

Thus any function $f = \sum_{i=1}^N a_i \kappa(\cdot, v_i) \in \mathcal{H}$ can be expressed as

$$\mathbf{f} = \begin{bmatrix} f(v_1) \\ \vdots \\ f(v_N) \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^N a_i \kappa(v_1, v_i) \\ \vdots \\ \sum_{i=1}^N a_i \kappa(v_N, v_i) \end{bmatrix} = \mathbf{K} \mathbf{a}$$

with $\mathbf{a} = (a_1, \dots, a_N)^T \in \mathbb{R}^N$, the norm of f can be expressed as [43, 52, 53]:

$$\|f\|_{\mathcal{H}}^2 = \sum_{i=1}^N \sum_{j=1}^N a_i a_j \kappa(v_i, v_j) = \mathbf{a}^T \mathbf{K} \mathbf{a}. \quad (2.14)$$

The theory of RKHS has been successfully applied to machine learning. Under certain conditions, the following Representer Theorem guarantees that a minimal problem can be solved in a lower-dimensional subspace.

Theorem 1 (Representer Theorem [52]). Let $\mathcal{V}$ be a non-empty set and $\mathcal{H}$ be an RKHS generated by a reproducing kernel on $\mathcal{V}$. For a subset $\mathcal{V}_0$ of $\mathcal{V}$, let

$$J(f) = \alpha(f) + \lambda \|f\|_{\mathcal{H}}^2,$$

where $\lambda > 0$ is a constant and $\alpha : \mathcal{H} \rightarrow \mathbb{R}$ is a function such that $\alpha(f)$ only depends on the values $\{f(v) \mid v \in \mathcal{V}_0\}$. If $f^* = \underset{f \in \mathcal{H}}{\text{argmin}} J(f)$, then the orthogonal projection of f^* on the space $\mathcal{H}_0 := \text{span}\{\kappa(\cdot, v) \mid v \in \mathcal{V}_0\}$, denoted by $g^* := P_{\mathcal{H}_0} f^*$, is the minimum point of $J(f)$ on the space $\mathcal{H}$. Thus

$$\min_{f \in \mathcal{H}} J(f) = \min_{g \in \mathcal{H}_0} J(g).$$

3. A fast algorithm for solving the BGSRP

In this section, we first give the analytic solution of the BGSRP. To reduce the computational cost, we propose a fast algorithm to solve the BGSRP based on the RKHS. The bright point is to skillfully construct a kernel and a corresponding RKHS, such that the smoothness term $\sigma(f)$ can be divided into the norm in the RKHS and the remaining term. Based on this result, the BGSRP can be solved in a lower-dimensional subspace and the computational cost reduces greatly.

3.1. The analytic solution

This section finds the analytic solution of the BGSRP (1.5). As described in Section 1, for an undirected, connected, weighted graph $\mathcal{G} = (\mathcal{V}, \mathbf{W})$, suppose $\mathcal{V}_0$ is a proper subset of $\mathcal{V}$ and $y : \mathcal{V}_0 \rightarrow \mathbb{R}$ is a function, our goal is to solve the BGSRP (1.5), i.e.,

$$\min_{f \in \mathcal{F}} \sum_{v \in \mathcal{V}_0} |f(v) - y(v)|^2 + \gamma \sigma(f)$$

for $\mathcal{F} := \mathbf{BL}_n(\mathcal{G})$, the bandlimited space defined by (2.9), and $\sigma(f) := \mathbf{f}^T \mathbf{L} \mathbf{f}$, where $\gamma > 0$ is a regularization parameter.

Denote by $\mathbf{E}_0$ the $\ell \times N$ matrix by extracting from $\mathbf{I}_N$ the rows with indices in $\mathcal{V}_0$, where $\ell := |\mathcal{V}_0|$ is the cardinality of $\mathcal{V}_0$. It is easy to see that

$$\sum_{v \in \mathcal{V}_0} |f(v) - y(v)|^2 = \|\mathbf{E}_0 \mathbf{f} - \mathbf{y}\|_2^2,$$

where $\|\cdot\|_2$ is the Euclidean norm. Then, the objective function of the BGSRP can be rewritten as

$$\begin{aligned} J(f) &= \|\mathbf{E}_0 \mathbf{f} - \mathbf{y}\|_2^2 + \gamma \mathbf{f}^T \mathbf{L} \mathbf{f} \\ &= \mathbf{f}^T (\mathbf{E}_0^T \mathbf{E}_0 + \gamma \mathbf{L}) \mathbf{f} - 2\mathbf{y}^T \mathbf{E}_0 \mathbf{f} + \|\mathbf{y}\|_2^2. \end{aligned}$$

Since any function in $\mathbf{BL}_n(\mathcal{G})$ can be expressed as

$$\mathbf{f} = \bar{\mathbf{U}}_n \mathbf{x}, \quad \text{where } \bar{\mathbf{U}}_n := (\mathbf{u}_1, \dots, \mathbf{u}_n), \quad \mathbf{x} \in \mathbb{R}^n,$$

and $\mathbf{u}_1, \dots, \mathbf{u}_n$ are the first n vectors of the graph Fourier basis, $J(f)$ can be further rewritten as the following quadratic form of $\mathbf{x}$:

$$J(f) = \mathbf{x}^T \bar{\mathbf{U}}_n^T (\mathbf{E}_0^T \mathbf{E}_0 + \gamma \mathbf{L}) \bar{\mathbf{U}}_n \mathbf{x} - 2\mathbf{y}^T \mathbf{E}_0 \bar{\mathbf{U}}_n \mathbf{x} + \|\mathbf{y}\|_2^2.$$

We claim that the coefficient matrix $\mathbf{C} := \bar{\mathbf{U}}_n^T (\mathbf{E}_0^T \mathbf{E}_0 + \gamma \mathbf{L}) \bar{\mathbf{U}}_n$ is a positive definite matrix. Actually, let $\mathbf{x}^T \mathbf{C} \mathbf{x} = 0$, then $\mathbf{x}^T \bar{\mathbf{U}}_n^T \mathbf{E}_0^T \mathbf{E}_0 \bar{\mathbf{U}}_n \mathbf{x} + \gamma \mathbf{x}^T \bar{\mathbf{U}}_n^T \mathbf{L} \bar{\mathbf{U}}_n \mathbf{x} = 0$. Since $\mathbf{x}^T \bar{\mathbf{U}}_n^T \mathbf{E}_0^T \mathbf{E}_0 \bar{\mathbf{U}}_n \mathbf{x} = \|\mathbf{E}_0 \bar{\mathbf{U}}_n \mathbf{x}\|_2^2 \geq 0$, we have $\mathbf{x}^T \bar{\mathbf{U}}_n^T \mathbf{L} \bar{\mathbf{U}}_n \mathbf{x} = 0$. Since $\bar{\mathbf{U}}_n^T \mathbf{L} \bar{\mathbf{U}}_n = \text{diag}(\lambda_1, \dots, \lambda_n)$ and $0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_n$, we have $x_2 = \dots = x_n = 0$, and consequently $\bar{\mathbf{U}}_n \mathbf{x} = x_1 \mathbf{u}_1$. Thus

$$0 = \mathbf{x}^T \mathbf{C} \mathbf{x} \geq x_1^2 \mathbf{u}_1^T \mathbf{E}_0^T \mathbf{E}_0 \mathbf{u}_1 = x_1^2 \|\mathbf{E}_0 \mathbf{u}_1\|_2^2.$$

Since each component of $\mathbf{u}_1$ is $1/\sqrt{N}$, we have $\|\mathbf{E}_0 \mathbf{u}_1\|_2^2 = \ell/N$ and consequently $x_1 = 0$, which proves the claim that $\mathbf{C}$ is a positive definite matrix. Hence the minimal problem

$$\min_{\mathbf{x} \in \mathbb{R}^n} (\mathbf{x}^T \mathbf{C} \mathbf{x} - 2\mathbf{y}^T \mathbf{E}_0 \bar{\mathbf{U}}_n \mathbf{x} + \|\mathbf{y}\|_2^2)$$

has an analytic solution $\mathbf{x}^* := \mathbf{C}^{-1} \bar{\mathbf{U}}_n^T \mathbf{E}_0^T \mathbf{y}$. Accordingly, the analytic solution of the BGSRP is

$$\mathbf{f}^* = \bar{\mathbf{U}}_n \mathbf{x}^* = \bar{\mathbf{U}}_n [\bar{\mathbf{U}}_n^T (\mathbf{E}_0^T \mathbf{E}_0 + \gamma \mathbf{L}) \bar{\mathbf{U}}_n]^{-1} \bar{\mathbf{U}}_n^T \mathbf{E}_0^T \mathbf{y}. \quad (3.15)$$

Let us consider the computational complexity of the analytic solution (3.15). Under the premise that $\bar{\mathbf{U}}_n$ is known, the computational complexity is $O(Nn + (n + \ell)n^2)$, as counted in the following. When $n = N$, the BGSRP degenerates into the problem (1.3) and the computational complexity is $O(N^3)$.

Below are the details of the computational complexity of (3.15). Let

$$\xi := \bar{\mathbf{U}}_n^T \mathbf{E}_0^T \mathbf{y}, \quad \mathbf{C} := (\mathbf{E}_0 \bar{\mathbf{U}}_n)^T (\mathbf{E}_0 \bar{\mathbf{U}}_n) + \gamma \bar{\mathbf{U}}_n^T \mathbf{L} \bar{\mathbf{U}}_n.$$

Then Eq. (3.15) can be rewritten as $\mathbf{f}^* = \bar{\mathbf{U}}_n \mathbf{C}^{-1} \xi$.

- Computing $\mathbf{C}^{-1}$: $\mathbf{E}_0 \bar{\mathbf{U}}_n$ can be obtained by directly extracting from $\bar{\mathbf{U}}_n$ those rows with indices in $\mathcal{V}_0$, the computational complexity can be ignored. The complexity for computing $(\mathbf{E}_0 \bar{\mathbf{U}}_n)^T (\mathbf{E}_0 \bar{\mathbf{U}}_n)$ is $O(n^2 \ell)$. Since $\bar{\mathbf{U}}_n^T \mathbf{L} \bar{\mathbf{U}}_n = \text{diag}(\lambda_1, \dots, \lambda_n)$, it requires only n flops to computing $\gamma \bar{\mathbf{U}}_n^T \mathbf{L} \bar{\mathbf{U}}_n$, which can be ignored. At last, $\mathbf{C}^{-1}$ can be calculated from $\mathbf{C}$ with $O(n^3)$ flops. Totally, the complexity for computing $\mathbf{C}^{-1}$ is $O(n^2 \ell + n^3)$.
- Computing ξ : $\xi = (\mathbf{E}_0 \bar{\mathbf{U}}_n)^T \mathbf{y}$ can be computed in $O(n \ell)$ flops.
- Computing $\mathbf{f}^*$: $\mathbf{f}^* = \bar{\mathbf{U}}_n (\mathbf{C}^{-1} \xi)$ can be obtained from $\bar{\mathbf{U}}_n$, $\mathbf{C}^{-1}$ and ξ in $O((N + n)n)$ flops.

In conclusion, the overall computational complexity of (3.15) is $O(n^2 \ell + n^3 + n \ell + (N + n)n)$, or equivalently, $O(Nn + (n + \ell)n^2)$.

3.2. An equivalent formulation of the BGSRP

As described above, we hope to represent $\sigma(f)$, the smoothness term of the objective function of the BGSRP, by the norm of an RKHS modulo a constant. To do so, this section firstly gives an equivalent formulation of the BGSRP. Let

$$\mathcal{H} := \text{span}\{\mathbf{u}_2, \dots, \mathbf{u}_n\}.$$

Then $\mathcal{F} = \text{span}\{\mathbf{u}_1\} \oplus \mathcal{H}$. For any $\mathbf{f} \in \mathcal{F}$, we have $\mathbf{f} = z\mathbf{u}_1 + \mathbf{g}$ with $\mathbf{g} := \sum_{i=2}^n x_i \mathbf{u}_i \in \mathcal{H}$ for some $z \in \mathbb{R}$ and $\mathbf{x} := (x_2, \dots, x_n)^T \in \mathbb{R}^{n-1}$. Hence

$$\sigma(\mathbf{f}) = \mathbf{f}^T \mathbf{L} \mathbf{f} = \sum_{i=2}^n \lambda_i x_i^2 = \mathbf{g}^T \mathbf{L} \mathbf{g},$$

and

$$\begin{aligned} \sum_{v \in \mathcal{V}_0} |f(v) - y(v)|^2 &= \sum_{v \in \mathcal{V}_0} |zu_1(v) + g(v) - y(v)|^2 = \|z\mathbf{E}_0 \mathbf{u}_1 + \mathbf{E}_0 \mathbf{g} - \mathbf{y}\|_2^2 \\ &= \|\mathbf{E}_0 \mathbf{u}_1\|_2^2 z^2 + 2z(\mathbf{E}_0 \mathbf{u}_1)^T (\mathbf{E}_0 \mathbf{g} - \mathbf{y}) + \|\mathbf{E}_0 \mathbf{g} - \mathbf{y}\|_2^2. \end{aligned} \quad (3.16)$$

Therefore, the objective function of problem (1.5) can be written as

$$J(z, \mathbf{g}) = \|\mathbf{E}_0 \mathbf{u}_1\|_2^2 z^2 + 2z(\mathbf{E}_0 \mathbf{u}_1)^T (\mathbf{E}_0 \mathbf{g} - \mathbf{y}) + \|\mathbf{E}_0 \mathbf{g} - \mathbf{y}\|_2^2 + \gamma \mathbf{g}^T \mathbf{L} \mathbf{g}.$$

If $\mathbf{g} \in \mathcal{H}$ is fixed, then $J(z, \mathbf{g})$ is a quadratic function of z , which reaches the minimum at

$$z(\mathbf{g}) = -\frac{(\mathbf{E}_0 \mathbf{u}_1)^T (\mathbf{E}_0 \mathbf{g} - \mathbf{y})}{\|\mathbf{E}_0 \mathbf{u}_1\|_2^2} \quad (3.17)$$

and the minimum is

$$J(z(\mathbf{g}), \mathbf{g}) = \|\mathbf{E}_0 \mathbf{g} - \mathbf{y}\|_2^2 - \left(\frac{(\mathbf{E}_0 \mathbf{u}_1)^T (\mathbf{E}_0 \mathbf{g} - \mathbf{y})}{\|\mathbf{E}_0 \mathbf{u}_1\|_2} \right)^2 + \gamma \mathbf{g}^T \mathbf{L} \mathbf{g}. \quad (3.18)$$

For the problem (1.5), we have the following theorem:

Theorem 2. *There exists $\mathbf{g}^* \in \mathcal{H}$ such that $J(z(\mathbf{g}^*), \mathbf{g}^*) \leq J(z(\mathbf{g}), \mathbf{g})$ for any $\mathbf{g} \in \mathcal{H}$. Furthermore,*

$$(z(\mathbf{g}^*), \mathbf{g}^*) = \underset{z \in \mathbb{R}, \mathbf{g} \in \mathcal{H}}{\text{argmin}} J(z, \mathbf{g}).$$

Proof. Let $\mathbf{g} \in \mathcal{H}$. Then $\mathbf{g} = \sum_{i=2}^n x_i \mathbf{u}_i = \mathbf{U}_n \mathbf{x}$, where $\mathbf{U}_n := (\mathbf{u}_2, \dots, \mathbf{u}_n)$ and $\mathbf{x} = (x_2, \dots, x_n)^T \in \mathbb{R}^{n-1}$. By (3.18) and the Cauchy-Schwartz inequality, we have that

$$\begin{aligned} J(z(\mathbf{g}), \mathbf{g}) &\geq \gamma \mathbf{g}^T \mathbf{L} \mathbf{g} = \gamma \sum_{i=2}^n \lambda_i x_i^2 \\ &\geq \gamma \lambda_2 \|\mathbf{x}\|_2^2 \rightarrow +\infty, \quad \|\mathbf{x}\|_2 \rightarrow +\infty. \end{aligned}$$

It is followed that $J(z(\mathbf{U}_n \mathbf{x}), \mathbf{U}_n \mathbf{x})$ reaches the minimum at some point $\mathbf{x}^*$ in $\mathbb{R}^{n-1}$. Consequently $\mathbf{g}^* := \mathbf{U}_n \mathbf{x}^*$ minimizes $J(z(\mathbf{g}), \mathbf{g})$ in $\mathcal{H}$.

Furthermore, for any $\mathbf{g} \in \mathcal{H}$ and $z \in \mathbb{R}$, we have

$$J(z(\mathbf{g}^*), \mathbf{g}^*) \leq J(z(\mathbf{g}), \mathbf{g}) \leq J(z, \mathbf{g}),$$

which proves that $(z(\mathbf{g}^*), \mathbf{g}^*) = \underset{z \in \mathbb{R}, \mathbf{g} \in \mathcal{H}}{\text{argmin}} J(z, \mathbf{g})$. $\square$

According to Theorem 2, the solution of problem (1.5) can be obtained by solving the following problem

$$\mathbf{g}^* := \underset{\mathbf{g} \in \mathcal{H}}{\text{argmin}} J(z(\mathbf{g}), \mathbf{g}), \quad (3.19)$$

and then calculating $\mathbf{f}^* = z(\mathbf{g}^*)\mathbf{u}_1 + \mathbf{g}^*$ by (3.17).

3.3. Converting the BGSRP in an RKHS

In this section, we show that the bandlimited space $\mathcal{F} = \mathbf{BL}_n(\mathcal{G})$ can be expressed by a reproducing kernel Hilbert space $\mathcal{H}$ and the smooth-

ness term $\mathbf{g}^T \mathbf{L} \mathbf{g}$ equals to the norm $\|\mathbf{g}\|_{\mathcal{H}}$ when the kernel function is properly chosen. Accordingly, based on Theorem 1, it is shown that the BGSRP (1.5) can be equivalently solved in a lower-dimensional subspace, which reduces the computational complexity greatly. Let

$$\mathbf{U}_n := (\mathbf{u}_2, \dots, \mathbf{u}_n), \quad \Phi_n := \text{diag}(\lambda_2^{-1}, \dots, \lambda_n^{-1}).$$

We construct a symmetric and positive semi-definite matrix of order N as follows:

$$\mathbf{K} := (K(i, j)) := \mathbf{U}_n \Phi_n \mathbf{U}_n^T.$$

Since $\Phi_n \mathbf{U}_n^T$ is row full rank ($\lambda_2^{-1} \mathbf{u}_2, \dots, \lambda_n^{-1} \mathbf{u}_n$ are linearly independent), we have that

$$\mathcal{H} = \text{span}\{\mathbf{u}_2, \dots, \mathbf{u}_n\} = \text{span}\{\kappa(\cdot, v) \mid v \in \mathcal{V}\},$$

where $\kappa: \mathcal{V} \times \mathcal{V} \rightarrow \mathbb{R}$ is a kernel function defined as $\kappa(v_i, v_j) := K(i, j)$, $i, j = 1, 2, \dots, N$. It means that $\mathcal{H}$ is an RKHS generated by the kernel function κ . Since

$$\mathbf{U}^T \mathbf{U}_n = \begin{pmatrix} \mathbf{u}_1^T \\ \vdots \\ \mathbf{u}_N^T \end{pmatrix} (\mathbf{u}_2, \dots, \mathbf{u}_n) = \begin{pmatrix} \mathbf{O} \\ \mathbf{I}_{n-1} \\ \mathbf{O} \end{pmatrix},$$

where the first row and the last $N - n$ rows of the last matrix are 0s and $\mathbf{I}_{n-1}$ is the identity matrix of order $n - 1$, it is deduced that

$$\mathbf{U}_n^T \mathbf{L} \mathbf{U}_n = \mathbf{U}_n^T \mathbf{U} \mathbf{L} \mathbf{U}^T \mathbf{U}_n = (\mathbf{O}, \mathbf{I}_{n-1}, \mathbf{O}) \Lambda \begin{pmatrix} \mathbf{O} \\ \mathbf{I}_{n-1} \\ \mathbf{O} \end{pmatrix} = \text{diag}(\lambda_2, \dots, \lambda_n),$$

and consequently

$$\mathbf{K} \mathbf{L} \mathbf{K} = \mathbf{U}_n \Phi_n \mathbf{U}_n^T \mathbf{L} \mathbf{U}_n \Phi_n \mathbf{U}_n^T = \mathbf{U}_n \text{diag}(\lambda_2^{-1}, \dots, \lambda_n^{-1}) \mathbf{U}_n^T = \mathbf{U}_n \Phi_n \mathbf{U}_n^T = \mathbf{K}.$$

Hence, the following critical equality

$$\mathbf{g}^T \mathbf{L} \mathbf{g} = \mathbf{x}^T \mathbf{K} \mathbf{L} \mathbf{K} \mathbf{x} = \mathbf{x}^T \mathbf{K} \mathbf{x} = \|\mathbf{g}\|_{\mathcal{H}}^2 \quad (3.20)$$

holds for any $\mathbf{g} = \mathbf{K} \mathbf{x} \in \mathcal{H}$ with $\mathbf{x} \in \mathbb{R}^N$. It is followed that the objective function of the problem (3.19) can be written as

$$J(z(\mathbf{g}), \mathbf{g}) = \alpha(\mathbf{g}) + \gamma \|\mathbf{g}\|_{\mathcal{H}}^2, \quad (3.21)$$

where

$$\alpha(\mathbf{g}) := \|\mathbf{E}_0 \mathbf{g} - \mathbf{y}\|_2^2 - \left(\frac{(\mathbf{E}_0 \mathbf{u}_1)^T (\mathbf{E}_0 \mathbf{g} - \mathbf{y})}{\|\mathbf{E}_0 \mathbf{u}_1\|_2} \right)^2. \quad (3.22)$$

Since $\alpha(\mathbf{g})$ is determined by $\mathbf{E}_0 \mathbf{g}$, i.e., the values of $\mathbf{g}$ on $\mathcal{V}_0$. According to Theorem 1, the problem (3.19) can be equivalently solved in the subspace $\mathcal{H}_0 := \text{span}\{\kappa(\cdot, v) \mid v \in \mathcal{V}_0\}$, that is,

$$\mathbf{g}^* := \underset{\mathbf{g} \in \mathcal{H}_0}{\text{argmin}} J(z(\mathbf{g}), \mathbf{g}). \quad (3.23)$$

Since $\dim \mathcal{H}_0 \leq \ell$, when ℓ is much smaller than N , the computational complexity for solving the problem (3.23) will be much lower than that of (3.19). Details on the computational complexity of problem (3.23) will be provided in the next section.

3.4. A fast algorithm and its computational complexity

Based on the conclusion of the last subsection, this subsection proposes a fast algorithm for solving the BGSRP (1.5). Let us firstly simplify the objective function defined by (3.21) and (3.22) of problem (3.23).

Since $\mathbf{u}_1 = \mathbf{1}_N / \sqrt{N}$, we have $\mathbf{E}_0 \mathbf{u}_1 = \mathbf{1}_\ell / \sqrt{N}$ and

$$\|\mathbf{E}_0 \mathbf{u}_1\|_2 = \sqrt{\ell/N}, \quad (\mathbf{E}_0 \mathbf{u}_1)(\mathbf{E}_0 \mathbf{u}_1)^T = \frac{1}{N} \mathbf{1}_{\ell \times \ell},$$

where $\mathbf{1}_{\ell \times \ell}$ is the matrix of order ℓ whose entries are all ones. Inserting these equalities into (3.22) we obtain

$$\alpha(\mathbf{g}) = \|\mathbf{E}_0 \mathbf{g} - \mathbf{y}\|_2^2 - \frac{N}{\ell} [(\mathbf{E}_0 \mathbf{u}_1)^T (\mathbf{E}_0 \mathbf{g} - \mathbf{y})]^2 = (\mathbf{E}_0 \mathbf{g} - \mathbf{y})^T \mathbf{A} (\mathbf{E}_0 \mathbf{g} - \mathbf{y}),$$

where

$$\mathbf{A} := \mathbf{I}_\ell - \frac{1}{\ell} \mathbf{1}_{\ell \times \ell}.$$

For any $\mathbf{g} \in \mathcal{H}_0$, we have $g(v) = \sum_{j=1}^\ell \xi_j \kappa(v, v_{\mathcal{V}_{0,j}})$, $\forall v \in \mathcal{V}$, where $v_{\mathcal{V}_{0,j}}, j = 1, 2, \dots, \ell$, represent those vertices in $\mathcal{V}_0$, i.e.,

$$\mathbf{g} = \mathbf{K} \mathbf{E}_0^T \boldsymbol{\xi}, \quad \text{where } \boldsymbol{\xi} := (\xi_1, \dots, \xi_\ell)^T \in \mathbb{R}^\ell,$$

which yields that

$$\|\mathbf{g}\|_{\mathcal{H}}^2 = (\mathbf{E}_0^T \boldsymbol{\xi})^T \mathbf{K} (\mathbf{E}_0^T \boldsymbol{\xi}) = \boldsymbol{\xi}^T \mathbf{E}_0 \mathbf{K} \mathbf{E}_0^T \boldsymbol{\xi},$$

$$\alpha(\mathbf{g}) = (\mathbf{E}_0 \mathbf{K} \mathbf{E}_0^T \boldsymbol{\xi} - \mathbf{y})^T \mathbf{A} (\mathbf{E}_0 \mathbf{K} \mathbf{E}_0^T \boldsymbol{\xi} - \mathbf{y}).$$

Inserting them into (3.21) we obtain that

$$\begin{aligned} J(z(\mathbf{g}), \mathbf{g}) &= \boldsymbol{\xi}^T (\mathbf{E}_0 \mathbf{K} \mathbf{E}_0^T \mathbf{A} \mathbf{E}_0 \mathbf{K} \mathbf{E}_0^T + \gamma \mathbf{E}_0 \mathbf{K} \mathbf{E}_0^T) \boldsymbol{\xi} - 2(\mathbf{E}_0 \mathbf{K} \mathbf{E}_0^T \mathbf{A} \mathbf{y})^T \boldsymbol{\xi} + \mathbf{y}^T \mathbf{A} \mathbf{y} \\ &= \boldsymbol{\xi}^T \mathbf{G} \boldsymbol{\xi} - 2\mathbf{d}^T \boldsymbol{\xi} + c, \end{aligned}$$

where

$$\begin{cases} \mathbf{G} := \mathbf{E}_0 (\mathbf{K} \mathbf{E}_0^T \mathbf{A} \mathbf{E}_0 \mathbf{K} + \gamma \mathbf{K}) \mathbf{E}_0^T, \\ \mathbf{d} := \mathbf{E}_0 \mathbf{K} \mathbf{E}_0^T \mathbf{A} \mathbf{y}, \\ c := \mathbf{y}^T \mathbf{A} \mathbf{y}. \end{cases} \quad (3.24)$$

Since $\mathbf{G}$ is a positive semi-definite matrix, we conclude that $\mathbf{g}^* = \mathbf{K} \mathbf{E}_0^T \boldsymbol{\xi}^*$ is the solution of problem (3.23) if and only if

$$\boldsymbol{\xi}^* = \underset{\boldsymbol{\xi} \in \mathbb{R}^\ell}{\text{argmin}} (\boldsymbol{\xi}^T \mathbf{G} \boldsymbol{\xi} - 2\mathbf{d}^T \boldsymbol{\xi} + c),$$

that is, $\boldsymbol{\xi}^*$ is the solution of the following linear system of equations:

$$\mathbf{G} \boldsymbol{\xi} = \mathbf{d}. \quad (3.25)$$

It is worth pointing out that since $\mathbf{G}$ is of order $\ell \times \ell$, when ℓ is much smaller than the bandwidth n , the computational complexity is greatly reduced comparing to the computational complexity of the analytic solution (3.15). Detailed estimation about the computational complexity will be given later.

For computational convenience, we further simplify the system (3.24). Let $\mathbf{B} := \mathbf{E}_0 \mathbf{U}_n$. Then, using $\mathbf{K} = \mathbf{U}_n \Phi_n \mathbf{U}_n^T$, (3.24) can be rewritten as

$$\begin{cases} \mathbf{G} = \mathbf{B} (\gamma \Phi_n + \Phi_n \mathbf{B}^T \mathbf{A} \mathbf{B} \Phi_n) \mathbf{B}^T, \\ \mathbf{d} = \mathbf{B} \Phi_n \mathbf{B}^T \mathbf{A} \mathbf{y}, \\ c = \mathbf{y}^T \mathbf{A} \mathbf{y}. \end{cases} \quad (3.26)$$

Once the problem (3.25) is solved, we can calculate $\mathbf{g}^* = \mathbf{K} \mathbf{E}_0^T \boldsymbol{\xi}^* = \mathbf{U}_n \Phi_n \mathbf{B}^T \boldsymbol{\xi}^*$ and $z(\mathbf{g}^*) = -\frac{\sqrt{N}}{\ell} \mathbf{1}_\ell^T (\mathbf{E}_0 \mathbf{g}^* - \mathbf{y})$. Finally, $\mathbf{f}^* = z(\mathbf{g}^*) \mathbf{u}_1 + \mathbf{g}^*$ is the solution of the BGSRP (1.5). We organize the above steps into the following Algorithm 1 for solving the BGSRP (1.5).

Let us estimate the computational complexity of the proposed Algorithm 1. For given $\mathbf{U}_n$ and $\{\lambda_k\}_{k=2}^n$,

- The first step requires $O(n)$ flops (floating point operations).
- The second step requires $O(n\ell^2)$ flops to compute $\mathbf{T}_\ell$ and $O(\ell^3)$ flops to compute $\mathbf{G}, \mathbf{d}$ and c . Thus the total complexity of this step is $O(\ell^2(n + \ell))$.
- The third step, it needs $O(\ell^3)$ flops to solve $\mathbf{G} \boldsymbol{\xi} = \mathbf{d}$.
- The fourth and fifth steps involve $O(Nn)$ flops.

In conclusion, the total computational complexity of the proposed algorithm is $O(Nn + (n + \ell)\ell^2)$, which is less than $O(Nn + (n + \ell)n^2)$, the complexity of the analytic solution (3.15), provided that $\ell \leq n$. Particularly, it can be seen that if $N + \ell^2 = o(n^2)$, the higher order infinitesimal of n^2 , then Algorithm 1 is much faster than (3.15).

Algorithm 1: Fast Algorithm for solving the BGSRP (1.5).

Input: A graph $\mathcal{G}$ with its Laplacian $\mathbf{L}$ and node set $\mathcal{V}$, a bandwidth n , the eigenvector matrix $\mathbf{U}_n = (\mathbf{u}_1, \dots, \mathbf{u}_n)$ and eigenvalues $\{\lambda_k\}_{k=2}^n$ of $\mathbf{L}$, a set $\mathcal{V}_0 \subset \mathcal{V}$ and a function $y: \mathcal{V}_0 \rightarrow \mathbb{R}$, a regularization parameter $\gamma > 0$.

1: Formulate

$$\Phi_n := \text{diag}(\lambda_2^{-1}, \dots, \lambda_n^{-1}).$$

2: Let $\ell := |\mathcal{V}_0|$ be the cardinality of $\mathcal{V}_0$ and $\mathbf{B} := \mathbf{U}_n(\mathcal{V}_0, :)$, where $\mathbf{U}_n(\mathcal{V}_0, :)$ denotes the $\ell \times n$ matrix consisting of all the rows whose indices are in $\mathcal{V}_0$. Denote $\mathbf{1}_\ell$ the ℓ -dimensional vector whose entries are all equal to 1. Compute $\mathbf{T}_\ell := \mathbf{B}\Phi_n\mathbf{B}^T$ and

$$\begin{cases} \mathbf{G} = \gamma\mathbf{T}_\ell + \mathbf{T}_\ell^2 - \frac{1}{\ell}(\mathbf{T}_\ell\mathbf{1}_\ell)(\mathbf{T}_\ell\mathbf{1}_\ell)^T, \\ \mathbf{d} = \mathbf{T}_\ell\mathbf{y} - \frac{1}{\ell}(\mathbf{1}_\ell^T\mathbf{y})(\mathbf{T}_\ell\mathbf{1}_\ell), \\ c = \mathbf{y}^T\mathbf{y} - \frac{1}{\ell}(\mathbf{1}_\ell^T\mathbf{y})^2. \end{cases}$$

3: Solve the linear system of equations $\mathbf{G}\xi = \mathbf{d}$ to get a solution ξ^* .

4: Compute

$$\mathbf{g}^* = \mathbf{U}_n\Phi_n\mathbf{B}^T\xi^* \quad \text{and} \quad z(\mathbf{g}^*) = -\frac{\sqrt{N}}{\ell}\mathbf{1}_\ell^T(\mathbf{g}^*(\mathcal{V}_0) - \mathbf{y}),$$

where $\mathbf{g}^*(\mathcal{V}_0)$ denotes the ℓ -dimensional vector consisting of all the components of $\mathbf{g}^*$ whose indexes are in $\mathcal{V}_0$.

5: Compute $\mathbf{f}^* = z(\mathbf{g}^*)\mathbf{u}_1 + \mathbf{g}^*$.

Output: The solution $\mathbf{f}^*$ of BGSRP (1.5).

At the end of this section, we give a remark on the solution of $\mathbf{G}\xi = \mathbf{d}$. When $\mathbf{G}$ is not a positive definite matrix, this equation has infinite solutions. However, we can show that the solution of problem (3.23) is unique.

Theorem 3. $\mathbf{G}$ is positive if and only if $\mathbf{E}_0\mathbf{U}_n$ is of row full rank. If $\mathbf{E}_0\mathbf{U}_n$ is not of row full rank, then the linear system of equations (3.25) has infinite solutions ξ and all the solutions produce the same $\mathbf{g} = \mathbf{K}\mathbf{E}_0^T\xi$.

Proof. Inserting $\mathbf{K} = \mathbf{U}_n\Phi_n\mathbf{U}_n^T$ into the expression of $\mathbf{G}$ in (3.24) we obtain that

$$\mathbf{G} := \mathbf{E}_0\mathbf{U}_n(\Phi_n\mathbf{U}_n^T\mathbf{E}_0^T\mathbf{A}\mathbf{E}_0\mathbf{U}_n\Phi_n + \gamma\Phi_n)\mathbf{U}_n^T\mathbf{E}_0^T.$$

Since $\Phi_n\mathbf{U}_n^T\mathbf{E}_0^T\mathbf{A}\mathbf{E}_0\mathbf{U}_n\Phi_n + \gamma\Phi_n$ is a positive definite matrix, it is easy to see that $\mathbf{G}$ is positive definite if and only if $\mathbf{U}_n^T\mathbf{E}_0^T\xi = \mathbf{0}$ implies $\xi = \mathbf{0}$, i.e., $\mathbf{E}_0\mathbf{U}_n$ is of row full rank.

If $\mathbf{E}_0\mathbf{U}_n$ is not of row full rank, then $\mathbf{G}$ is not either and the general solution of (3.25) can be expressed as

$$\xi_d + \xi,$$

where ξ_d is a particular solution of (3.25) and ξ satisfies $\mathbf{G}\xi = \mathbf{0}$. As discussed above, $\mathbf{G}\xi = \mathbf{0}$ implies that $\mathbf{U}_n^T\mathbf{E}_0^T\xi = \mathbf{0}$. Hence

$$\mathbf{g} := \mathbf{K}\mathbf{E}_0^T(\xi_d + \xi) = \mathbf{K}\mathbf{E}_0^T\xi_d + \mathbf{K}\mathbf{E}_0^T\xi = \mathbf{K}\mathbf{E}_0^T\xi_d,$$

which shows $\mathbf{g}$ produced by $\xi_d + \xi$ is equal to that produced by ξ_d . The proof ends. $\square$

4. Performance analysis and experimental comparison

In this section, experiments are conducted to study the performance of Algorithm 1. We consider the following three kinds of artificial graphs in GSPBOX and real datasets of recommendation system, respectively. GSPBOX is a MATLAB toolbox oriented to graph signal processing [54] and three kinds of graphs are described as follows.

- (a) ‘sensor’ graph, a network with assigned number of vertices and edge weights assigned with a threshold Gaussian kernel;
- (b) ‘bunny’ graph, a point cloud of the Stanford bunny with 2503 vertices and nearest-neighbor weights;

- (c) ‘minnesota’ graph, a Minnesota road network with 2642 vertices and all edge weights are equal to 1.

In all the following experiments, the mean absolute error ‘MAE’ between the original signal $\mathbf{f}$ and the reconstructed signal $\mathbf{g}$ is calculated by $\text{MAE} := \frac{1}{N} \sum_{i=1}^N |f_i - g_i|$. ‘TIME’ represents the seconds taken by the algorithm. The codes are available at <https://github.com/mazhangq/Code-for-BGSRP>.

4.1. An explicit example of signal recovery on ‘Minnesota’ graph

Firstly, we conduct a graph signal recovery example on ‘minnesota’ graph. We pre-compute the eigendecomposition of graph Laplacian by using GSPBOX. An original bandlimited graph signal $\mathbf{f}$ is generated by $\mathbf{f} := \sum_{k=1}^{10} c_k \mathbf{u}_k$, where the non-zero graph Fourier coefficients $\{c_k\}_{k=1}^{10}$ of the signal are randomly generated from the interval (0, 1), and $\{\mathbf{u}_k\}_{k=1}^{10}$ are the first 10 eigenvectors of the graph Laplacian. We randomly select 10% of the vertices on the graph as labeled vertices, thus the signal to be reconstructed in the experiment only retains the labeled values, and the signal values at the unlabeled vertices are set to 0. The partial signal to be reconstructed is shown in the left subfigure of Fig. 1. The blue and red lines represent the magnitudes of the positive and negative components of the signal, respectively. After selecting parameters $\gamma = 0.1$ and $n = 500$ in Algorithm 1, we display the reconstructed signal in the right subfigure of Fig. 1.

4.2. Affections of the bandwidth n

As studied in Section 3.4, under the premise that $\mathbf{U}_n$ and $\mathbf{A}_n$ are known, the BGSRP (1.5) can be solved by Algorithm 1 in $O(Nn + (n + \ell)\ell^2)$ flops. This means that the bandwidth n used in Algorithm 1 affects the computational complexity. An empirical bandwidth choice is to set the bandwidth n in Algorithm 1 equal to the true bandwidth n_0 of the graph signal to be reconstructed. However, the true bandwidth n_0 is usually unknown in practical applications. In order to find knowledge on the choice of n , we investigate how the value of n affects the reconstruction error by Algorithm 1.

Before experiments, we need to choose the regularization parameter γ in (1.5), which is used to balance the weight between the error term and the signal smoothness. There are no general techniques on this issue in the theory of optimization. We conduct numerical experiments for our algorithm with many different parameters and find that it does not produce essential impact on the final reconstruction error and calculation time consuming. Without loss of generality, in the following experiments involved in this paper, we fix the regularization parameter $\gamma = 0.1$ empirically.

We generate four original signals on the graph ‘sensor’ with $N = 400$, marked by ‘f1’, ‘f2’, ‘f3’ and ‘f4’ with four different bandwidths $n_0 = Nr_0$ for $r_0 = 0.2, 0.3, 0.4$ and 0.5 . Then we sample these signals randomly to get the labeled data $\{y(v)|v \in \mathcal{V}_0\}$ with $\ell = N \times 30\%$. By setting $n = Nr$ for $r = 0.1, 0.2, \dots, 1.0$, we use Algorithm 1 to reconstruct these signals. Finally, the MAEs are displayed in the middle subfigure of Fig. 2, in which the x -axis represents the bandwidth ratio $r := n/N$. Similar experiments are conducted on the graph ‘bunny’ of $N = 2503$ vertices with the same parameters r_0 and r . The results are displayed in the middle subfigure of Fig. 3.

It is seen that, as r increases from 10% to 100% the errors decrease rapidly at the beginning and then become stable after $r \geq r_0$. Besides, we find an interesting phenomenon that the reconstruction MAE achieves the minimum when n used in Algorithm 1 is close to the true bandwidth n_0 of the original signal.

It can be roughly explained as follows: Let $\mathbf{f}_0 = \sum_{i=1}^{n_0} c_i \mathbf{u}_i \in \mathbf{BL}_{n_0}(\mathcal{G})$ be the original signal and $\mathbf{f}^* := \sum_{i=1}^n c_i^* \mathbf{u}_i \in \mathbf{BL}_n(\mathcal{G})$ be the solution of the BGSRP. Then for $n < n_0$ we have that

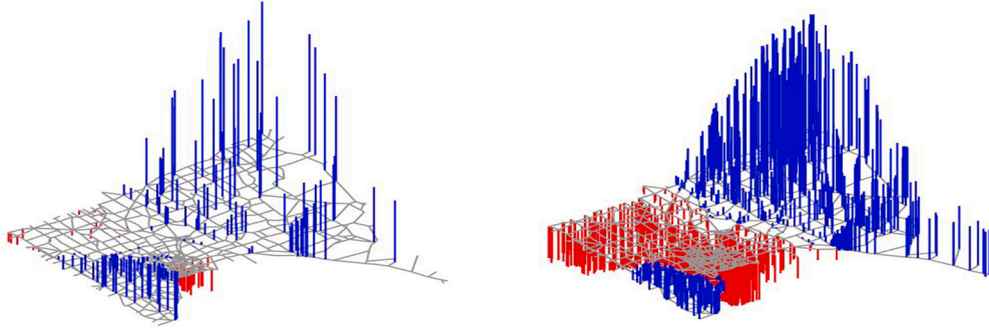


Fig. 1. An explicit example of signal reconstruction on ‘minnesota’ graph. Left: The partial signal to be reconstructed; Right: The signal reconstructed by Algorithm 1.

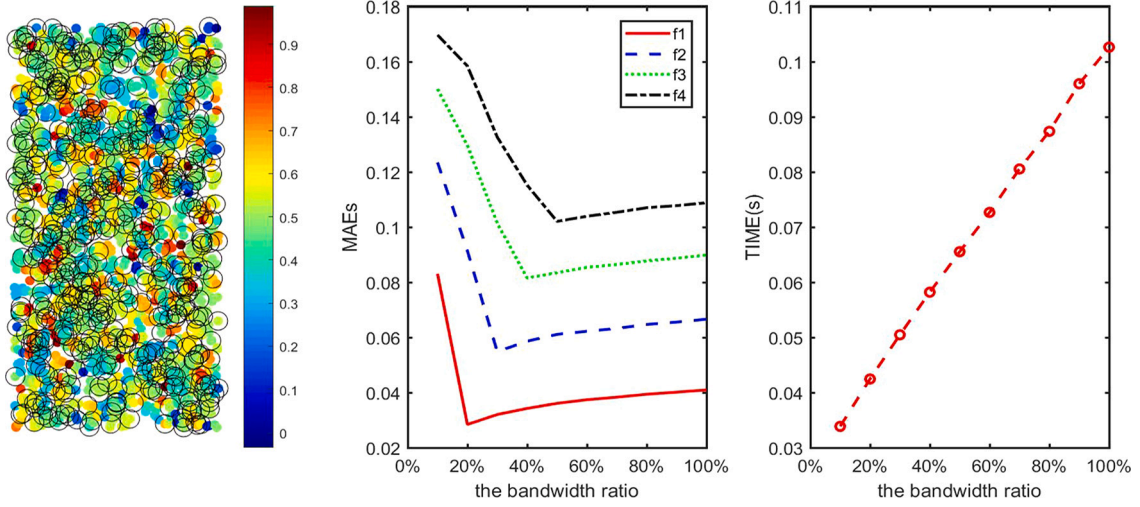


Fig. 2. Left: a bandlimited signal defined on the graph ‘sensor’; Middle: MAEs as functions of bandwidth ratio $r = n/N$; Right: Seconds used for reconstructing the signal at different bandwidth ratio.

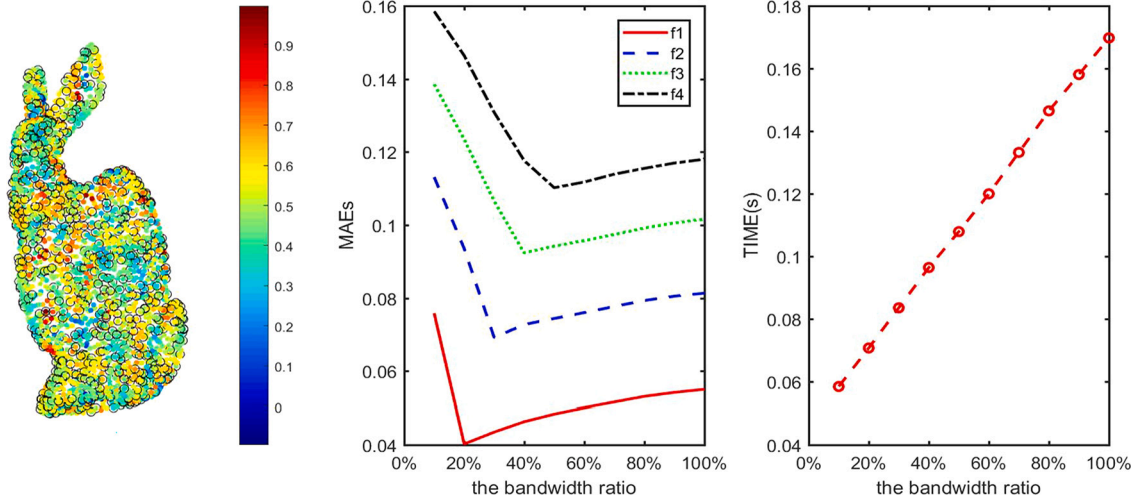


Fig. 3. Left: a bandlimited signal defined on the graph ‘bunny’; Middle: MAEs as functions of bandwidth ratio $r = n/N$; Right: Seconds used for reconstructing the signal at different bandwidth ratio.

$$\|\mathbf{f}_0 - \mathbf{f}^*\|_2^2 = \sum_{i=1}^n |c_i - c_i^*|^2 + \sum_{i=n+1}^{n_0} |c_i|^2 \geq \sum_{i=n+1}^{n_0} |c_i|^2,$$

which decreases monotonically as n increases from 1 to n_0 . Similarly, for $n > n_0$ we have that

$$\|\mathbf{f}_0 - \mathbf{f}^*\|_2^2 = \sum_{i=1}^{n_0} |c_i - c_i^*|^2 + \sum_{i=n_0+1}^n |c_i^*|^2,$$

whose second term is close to 0 since $c_i^* \approx 0$ for $i = n_0 + 1, \dots, n$.

This result shows that the best choice of n is n_0 , which achieves the minimal error while making the operation cost the least. It is a challenging problem to estimate the bandwidth of a signal from a few samples. In practice we suggest choosing $n = Nr$ with bandwidth ratio between 0.2 and 0.5.

The complexity $O(Nn + (n + \ell)\ell^2)$ of Algorithm 1 depends linearly on n . To verify this theoretic result, we generate 1000 original signals

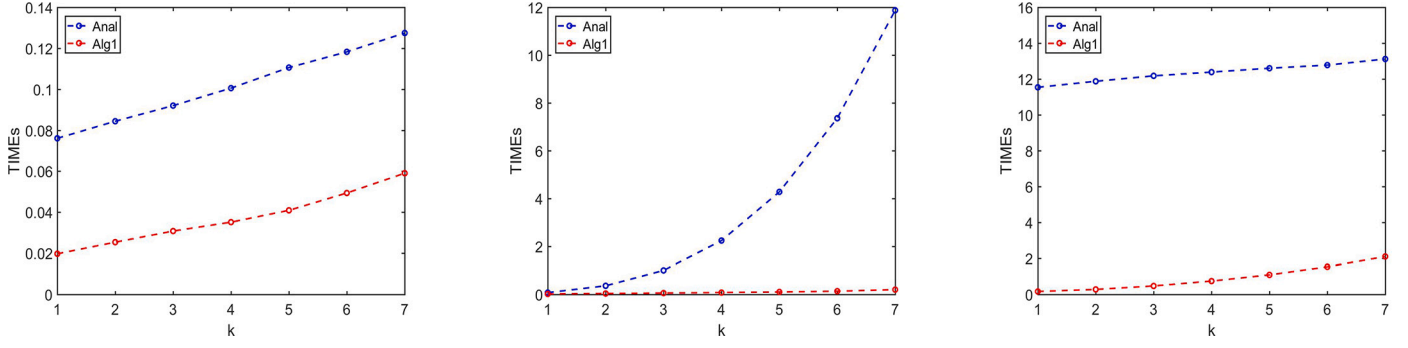


Fig. 4. Computational complexity on N , n and ℓ of Algorithm 1 and analytic solution (3.15). Left: TIMES for reconstructing by varying the graph size in the range $N = 10000k + 10000$ for $k = 1, \dots, 7$; Middle: TIMES for reconstructing by varying the bandwidth in the range $n = 1000k$ for $k = 1, \dots, 7$; Right: TIMES for reconstructing by varying the number of known labeled points in the range $\ell = 300k - 100$ for $k = 1, \dots, 7$.

on the graphs ‘sensor’ and ‘bunny’ with random bandwidths and then sample these signals with $\ell = N \times 30\%$. We reconstruct these signals by Algorithm 1 with various bandwidths $n = Nr$ for $r = 0.1, 0.2, \dots, 1.0$. The average calculation TIMES for reconstructing these 1000 signals are depicted in the right subfigures of Figs. 2 and 3, in which the x-axis is the bandwidth ratio r . It is shown that, the average calculation TIME grows linearly as n increases.

4.3. Computational complexity of Algorithm 1

The experiments in this part are carried out around the computational complexity of Algorithm 1.

Firstly, we verify by experiments the complexity of Algorithm 1 and the analytic solution solved by formula (3.15) when U_n and $\{\lambda_k\}_{k=2}^n$ are known a priori, thus the cost of the eigendecomposition is not taken into counting. We present the results in three scenarios:

$$\begin{cases} (a) \ell = 200, n = 1000, N = 10000(k + 1), \\ (b) \ell = 200, N = 20000, n = 1000k, & k = 1, \dots, 7. \\ (c) n = 7000, N = 20000, \ell = 300k - 100, \end{cases}$$

In the left subfigure of Fig. 4, we conduct the experiment on 7 ‘sensor’ graphs with different vertex numbers $N = 10000(k + 1)$ for $k = 1, \dots, 7$. For each graph, we generate randomly 100 original bandlimited signals with the bandwidth $n_0 = 1000$ and sample each signal randomly with $\ell = 200$. Each graph signal is reconstructed by Algorithm 1 and the analytic solution (3.15) with $n = 1000$. The average computational TIMES for reconstructing these 100 signals by Algorithm 1 and (3.15) are depicted as the red and blue dotted lines, marked as ‘Alg1’ and ‘Anal’, respectively. It is seen from the left subfigure that the average reconstruction TIMES of Algorithm 1 and (3.15) are linearly growing with the increasing of vertex numbers, which coincides with the theoretic complexity results. Meanwhile, the reconstruction time by Algorithm 1 is less than the time consumed by the analytic solution in all of cases.

In the middle subfigure of Fig. 4, we fix the graph size with $N = 20000$ vertices and the known labeled vertices with $\ell = 200$. We generate 100 original bandlimited signals per each bandwidth n_0 , by varying in the range $n_0 = 1000k$, $k = 1, \dots, 7$. Each graph signal is reconstructed by Algorithm 1 and the analytic solution (3.15) with $n = n_0$. Similarly, the average computational TIMES for reconstructing these 100 signals are depicted. It is seen from the middle subfigure that the average reconstruction TIME of (3.15) increases rapidly with the increase of n , while the average reconstruction TIME of Algorithm 1 has a slightly increase. The main reason for the difference lies in the fact that the analytic solution needs to calculate an n -dimensional linear system, its total computational complexity is $O(Nn + (n + \ell)n^2)$, while Algorithm 1 finds the solution in a lower-dimensional space, its complexity is mainly related to the size ℓ of known labeled vertices, i.e. $O(Nn + (n + \ell)\ell^2)$.

Along with increase of n , the calculation time difference between these two algorithms becomes more obvious.

In the right subfigure of Fig. 4, we fix the graph size with $N = 20000$ and the bandwidth with $n = 7000$. Let the size of known labeled sets change in the range $\ell = 300k - 100$, $k = 1, \dots, 7$, and for each case we also generate 100 original bandlimited signals with the bandwidth $n_0 = n$. Similarly, the average reconstruction TIMES of Algorithm 1 and (3.15) are depicted. It is seen from the right subfigure that the relationship between the reconstruction time of the analytic solution and ℓ is not obvious, when $\ell = 200$ or $\ell = 2000$, the difference of reconstruction time by analytic solution is only about 1 second, less than 10% of the total reconstruction time. On the contrary, the computational time of Algorithm 1 forms a significant increasing function of ℓ . From $\ell = 2000$ to $\ell = 200$, the reconstruction time drops from 2 seconds to less than 0.2 seconds, decreasing amplitude is more than 90%. This also coincides with the theoretic complexity of Algorithm 1, which has obvious advantages in dealing with small-scale known labeled set problems.

4.4. Experiments on MovieLens-100k

In this subsection, we turn to the experiments on the real world data in the recommendation system. Recommendation system (RS) has been extensively applied in electronic commerce. Some well-known e-commerce platforms, including Amazon, YouTube, Netflix and Yahoo, are gradually equipped with RSs. The core problem of RS is to predict the ‘rating’ of a user would give to an item according to the ratings of a few users to a few items.

One of the most commonly used data set in RS is the MovieLens-100k [55]. It consists of 100,000 ratings from 943 users on 1682 movies, the users and movies are denoted by U and I . The ratings are integer values between 1 and 5. The data set has 5-fold cross validation by partitioning the data set into 5 random subsets, named as $u1.base - u5.base$ and $u1.test - u5.test$. Each pair of $u*.base$ and $u*.test$ is a 80%/20% split of the whole data set, $u*.base$ for training and $u*.test$ for test.

In the recommendation system, we first need to solve two problems. The first is data preprocessing. Since there are few known ratings in the data set MovieLens-100k, for some movies, if the user’s ratings are too less, it is difficult to make an accurate prediction from the known ratings. Therefore, we filter out the movies when rating ratio is less than 3%. The second is to construct the graph adjacent matrix. The adjacency matrix should reflect the degree of consensus between different users on the same item, which is crucial to the accuracy of subsequent predictions. In the experiments, we consider the most used proximity metrics between users to construct the weighted adjacency matrix W . The cosine (COS) similarity is computed as the cosine of the angle between users u and v :

Table 1

Global MAEs and TIMEs for different methods on the sets u1~u5.

	MAEs					
	Ours	Tik	Var	Samp	Pocs	Psd
u1	0.807	0.807	0.807	0.807	0.841	0.790
u2	0.800	0.800	0.799	0.801	0.827	0.775
u3	0.794	0.795	0.797	0.794	0.814	0.766
u4	0.795	0.801	0.796	0.807	0.819	0.764
u5	0.801	0.802	0.800	0.814	0.820	0.768

	TIMEs					
	Ours	Tik	Var	Samp	Pocs	Psd
u1	1.00	30.23	39.05	30.18	34.80	90.64
u2	0.89	30.28	39.01	30.50	33.14	90.22
u3	0.83	30.56	39.43	30.83	32.80	91.42
u4	0.81	30.57	39.24	31.03	32.77	90.67
u5	0.80	30.91	39.17	30.59	32.75	92.09

$$W(u, v) = \frac{\sum_{i \in I} r(u, i) r(v, i)}{\sqrt{\sum_{i \in I} r^2(u, i)} \sqrt{\sum_{i \in I} r^2(v, i)}}. \quad (4.27)$$

Now we begin the 5-fold cross validation on the subsets $u1.base - u5.base$ and $u1.test - u5.test$. For each cross validation round, we calculate the COS similarity to construct the weighted adjacency matrix $\mathbf{W}$ and the graph Laplacian $\mathbf{L}$. Our objective is to fill in all missing movie ratings, i.e. run the reconstruction problem 1682 times. When all the ratings are filled in, we compare the 20000 ratings in the test set with the estimated results at the corresponding locations. Our proposed algorithm ('Ours') will be compared with the benchmark algorithms: 'Tik' [49], 'Var' [48], 'Samp' [30], 'Pocs' [32] and 'Psd' [39]. The MAEs and TIMEs of the experiments on the five data sets u1~u5 are listed in Table 1. In terms of MAE, the first five methods in Table 1 present similar reconstruction accuracy, the method 'Psd' has better prediction effect, but consumes the most calculation time at the same time. In terms of reconstruction time, the experimental results show that 'Ours' method presents obvious advantage in the efficiency of calculation.

5. Conclusion

In this paper, we focus on the bandlimited graph signal recovery problem (BGRSP). We skillfully construct a kernel function, such that the smoothness item is equivalent to the norm in an RKHS module a constant. By using the Representer Theorem in the RKHS, the solution of the model can be found in a lower-dimensional subspace. Compared with the computational complexity of the analytical solution in Section 3.1, the proposed Algorithm 1 reduces the computational complexity from $O(Nn + (n + \ell)n^2)$ to $O(Nn + (n + \ell)\ell^2)$ provided that ℓ is less than n . Particularly, if $N + \ell^2 = o(n^2)$, then Algorithm 1 is much faster than the computation of the analytic solution. Experimental results verify the complexity analysis of our algorithm and demonstrate the efficiency of the calculation.

The strategy of applying the Representer Theorem in the RKHS with inadequate conditions is powerful and is possible to be extended to deal with other optimization models.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

This research was supported by National Natural Science Foundation of China (Nos. 12171488, 11501377), Guangdong Province Key Grant (No. 2020B1212060032), Guangdong Provincial Key Laboratory (No. 2023B1212060076).

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